Project Design Phase

# Problem – Solution Fit Template

*Smart Library Request Workflow System*

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|---|---|
| Date | 08 April 2026 |
| Team ID | SWTID-2026-2921 |
| Project Name | Smart library request workflow in servicenow |
| Maximum Marks | 2 Marks |

## Problem – Solution Fit: Overview

The Problem-Solution Fit confirms that a clearly identified problem exists for a defined customer segment, and that the proposed solution meaningfully resolves it. For the Smart Library Request Workflow System, this means validating that current library processes cause real friction for students, researchers, and library staff — and that a digital workflow system genuinely addresses those friction points.

### Purpose of this Template

- Identify real pain points in the current library book request and borrowing process.
- Match each problem to a specific customer segment experiencing it.
- Map existing workarounds and explain why they fall short.
- Propose targeted solutions that integrate into users' existing behaviours.
- Define measurable key metrics to validate that the solution achieves problem-solution fit.

## Problem – Solution Fit Canvas

The canvas below maps each identified problem to the affected customer segment, evaluates existing alternatives, proposes a specific solution, and defines the success metric that will confirm fit.

| Problem | Customer Segment | Existing Alternatives | Proposed Solution | Key Metrics |
|---|---|---|---|---|
| Students cannot request books online; must visit the library in person. | **University Students** | Manual paper-based forms or email requests with slow turnaround. | Online self-service portal with real-time availability check and request submission. | Request processing time reduced from 48 hrs to under 2 hrs. |

| Problem | Customer Segment | Existing Alternatives | Proposed Solution | Key Metrics |
|---|---|---|---|---|
| Librarians manually track hundreds of requests with no unified system. | **Library Staff / Librarians** | Spreadsheets and verbal communication; error-prone and time-consuming. | Centralised dashboard for librarians to approve, assign, and track all requests. | 90% reduction in manual tracking errors; staff productivity up by 40%. |
| Researchers have no visibility on inter-library loan (ILL) status. | **Researchers & Faculty** | Phone calls or email follow-ups; no tracking or ETA provided. | Automated ILL request module with real-time status updates and SMS/email alerts. | ILL request visibility improved to 100%; average wait-time communicated upfront. |
| Users are unaware of new arrivals or availability of reserved books. | **General Library Members** | Notice boards and weekly newsletters; outdated and not personalised. | Push notification system alerting users when reserved books are available or new arrivals match interests. | Book pickup rate for reserved items increases from 55% to 92%. |

## Smart Library Request Workflow – End-to-End Process

The table below outlines the complete seven-stage workflow that the Smart Library Request system will implement, from the moment a user submits a request to the final return and feedback capture.

| # | Stage | Activity | Stakeholder | Output |
|---|---|---|---|---|
| **1** | **Request Submission** | User submits a book/resource request via web portal or mobile app, selecting title, format and urgency. | Student / Researcher | Request ticket created |
| **2** | **Availability Check** | System auto-checks catalogue inventory and flags if item is available, on loan, or requires ILL. | System (AI) | Availability status |
| **3** | **Librarian Review** | Librarian reviews edge cases: special collections, restricted access, or ILL escalations. | Librarian | Approval / Escalation |
| **4** | **Fulfilment & Assignment** | Approved requests are assigned to physical copies; ILL requests are forwarded to partner libraries. | Librarian / System | Assignment record |

| # | Stage | Activity | Stakeholder | Output |
|---|---|---|---|---|
| 5 | **Notification** | Requester receives real-time alert (email/SMS) with pickup location, time slot, and due date. | System | Notification sent |
| 6 | **Pickup & Loan** | User collects the book; system logs loan, updates inventory, and starts due-date countdown. | Student / Librarian | Loan record updated |
| 7 | **Return & Feedback** | Book returned; system closes the ticket; user prompted for satisfaction feedback. | Student / System | Feedback captured |

## Fit Validation Framework

The following framework assesses the proposed Smart Library solution across four critical fit dimensions: Desirability, Feasibility, Viability, and Scalability. Each dimension is validated through a specific method.

| Fit Dimension | Problem Signal | Solution Response | Validation Method |
|---|---|---|---|
| **Desirability** | Users want faster, contactless book access | Self-service digital request portal | User surveys & pilot testing |
| **Feasibility** | Library has existing catalogue database | API integration with LMS (Library Mgmt System) | Technical prototype sprint |
| **Viability** | Operational cost of manual processing is high | Automation reduces staff workload by ~40% | Cost-benefit analysis |
| **Scalability** | Multi-campus and inter-library demand is growing | Cloud-based platform supports multi-branch expansion | Load testing & roadmap review |

### References


1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe